A-A-2551
February 6, 1988

COMMERCIAL ITEM DESCRIPTION

DRILL, HAND

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification GGG-D-671, Type I, Class 1, Style B.

This commercial item description covers hollow-handled, single speed, and non-ratchet hand drills.

Salient characteristics.

Drills, hand. Drills shall consist essentially of a frame, spindle, chuck, crank, handles, side grip or knob, and driving, idler, and pinion gears. The drills shall have an overall length of not less than 11-1/2 inches nor more than 16 inches with a maximum weight of 5 pounds.

Material. The material used in the manufacture of the drills shall conform to the requirements hereinafter specified. The material shall be free from defects and imperfections that may affect the serviceability of the finished products.

Coating. The housing or cast portion of the frame and the nonmachined surfaces of the driving gear shall be enameled or coated with one of the stated finishes. Unless otherwise specified, the chuck crank arm and ferrules shall be plated with zinc. Other exposed bare metal surfaces shall be plated with zinc. When zinc is restricted by Government order, nickel plate, chromium plate, hot phosphating, or oxide coating shall be used.

Frame. The frame shall be of box or "H" design and of malleable iron, steel or other metal for the purpose, and of strong and durable construction.

Spindle. The spindle shall be finished steel and round within 0.001 inch. A ball type bearing or a single ball shall be provided to absorb the axial thrust and minimize spindle binding.

Chuck. The chuck shall be steel with three hardened jaws capable of gripping the shanks of drill bits by hand tightening only. The jaws shall show a Rockwell hardness value of not less than 53 nor more than 62 on the C scale, or the jaws may be casehardened to show a hardness value of not less than 87 nor more than 91 on the Rockwell 15N scale (superficial). The Rockwell Hardness Test shall be performed in accordance with ASTM E18.

Crank arm. The crank arm shall be steel. The arm of the crank, measured from the center of the pivot to the center of the handle, shall be not less than 3 nor more than 4 inches for the 3/8-inch capacity drills and not less than 2 nor more than 3-1/2 inches for the 1/4-inch capacity drills.

Handles and knobs. All handles and knobs shall be water-resistant treated hardwood, sized and formed and sanded to a smooth finish, or of phenolic, polypropylene, or ABS plastic.

Drill point set. Straight shank jobbers length twist drills shall conform to Table I in increments of 1/64 inch and shall be furnished in the hollow-end gripping handle. Drill points shall be made of steel having a 0.70 to 0.80 percent carbon content and shall show a Rockwell hardness value of not less than 48 nor more than 54 on the C scale.

DISTRIBUTION STATEMENT A. Approved for Public Release; Distribution is Unlimited.

FSC 5110

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NOTES

Table I.

Drill diameter Inch	Overall length ±1/8 Inches	Flute length ±1/8 Inches
1/16	1-7/8	7/8
5/64	2	1
3/32	2-1/4	1-1/4
7/64	2-5/8	1-1/2
1/8	2-3/4	1-5/8
9/64	2-7/8	1-3/4
5/32	3-1/8	2
11/64	3-1/4	2-1/8

Driving, idler, and pinion gears. All gears shall have beveled teeth free from any imperfections, burrs, or defects which may affect their serviceability. The face width of the gear teeth shall be not less than 23/64 inch (minimum) for the 3/8 inch capacity drills and 15/64 inch (minimum) for the 1/4-inch capacity drills with a maximum backlash of .020 inch. The driving gear shall be of malleable or gray iron, or zinc alloy conforming to SAE 903, with means provided for firmly attaching the crank. The idler gear and driven gear (pinion) shall be of steel or zinc alloy conforming to SAE 903.

Drill capacity. The hand drills shall be furnished with a drill capacity of either 0 up to and including 1/4 inch, or 0 up to and including 3/8 inch, as specified in the purchase order.

Identification marking. Each drill shall be etched or stamped in a permanent and legible manner with the manufacturer's name or identifying symbol of such known character that the source may be readily identifiable, part number, and country of origin.

Workmanship. All drills shall be free from imperfections which may affect their safety, appearance, serviceability, or durability.

QUALITY ASSURANCE PROVISIONS

Sampling. Sampling for inspection and testing shall be in accordance with MIL-STD-105, Sampling Procedures and Tables for inspection by Attributes, except where otherwise specified herein. The lot size shall consist of all drills offered for acceptance at one time. The sample unit is one drill.

Visual and dimensional inspection. Each drill in the sample lot shall be examined for workmanship, marking, and dimensional requirements. The inspection level shall be I with an acceptable quality level (AQL) of 4.0 defects per hundred units.

Testing of end item. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

Drilling test. The drills shall be subjected to drilling a cold-rolled steel (1010 grade minimum) sheet or plate in a vertical position. A vise or clamp shall be used to assure perpendicularity between the drill and the steel plate. The steel sheet or plate shall be not less than 1/8 nor more than 3/16 inch thick. A maximum-size drill bit within the capacity range of the drill shall be inserted in the chuck jaws and the chuck tightened by hand. The crank shall be turned by hand with a forward stroke at a speed not less than 80 revolutions per minute (rpm) and not greater than 100 rpm with a 40 pound + 5 pound pressure, applied on large end of the handle. The longitudinal axis of the drill shall be held at right angles to the steel plate. When the drill is about to pierce the opposite side of the plate, the 40 pound load shall be released and normal pressure applied on the handle until the drill has pierced entirely through the plate. Each individual drill subjected to the test shall make at least one such drilling operation to determine that the drill is in compliance with the requirements of this description. The drill and its component members shall show no mechanical defects and shall not fail, fracture, take permanent set or show any evidence of binding or undue wear.

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Concentricity test. The frame of the drill shall be gripped in a bench vise or by other means, with the chuck in a horizontal plane. A test rod of any diameter within the capacity range of the chuck to be tested, shall be inserted in the drill chuck, and the chuck or drill sleeve tightened by hand. The test rod shall project 2.0 inches from the jaws when inserted to the full depth of the drill chuck. This test rod shall not run out more than .0075 of an inch (average of three chuckings of the test rod for each sample drill tested) total indicator reading in 360 degrees rotation. The entire exposed length of the test rod shall be within the concentricity requirements.

Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing supplier's employees to undue hazards from the recovered materials.

Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Notes.

The issue of referenced documents in effect on date of invitation for bid or request for proposal shall be used to determine compliance with the stated requirements. Sources for obtaining referenced documents may be determined by contacting the contracting officer.

SAE standards are available from Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.

ASTM standards are available from American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

Military Standards are available from Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

MILITARY INTERESTS:

Military Coordinating Activity
Army - GL

Custodians
Army - GL
Air Force - 99

Review Activities
Air Force - 84

PREPARING ACTIVITY:

GSA/FSS

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